



OPERATING SYSTEM DESIGN

S Thenmozhi

Department of Computer Applications

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Memory Management and File System Management

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Allocating Virtual Memory

1. Check swap usage

free -h

```
thenmozhi@thenmozhiVM:~/Desktop$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	2.4Gi	806Mi	466Mi	9.0Mi	1.2Gi	1.5Gi
Swap:	1.1Gi	0B	1.1Gi			

Allocating Virtual Memory

1. To check existing Swap space

```
sudo swapon -show
```

2. To create a swapfile

```
sudo fallocate -l 2G /var/swapfile
```

3. To set swap permissions

```
sudo chmod 600 /var/swapfile
```

4. Make the file swappable

```
sudo mkswap /var/swapfile
```

5. Activate the swapfile

```
sudo swapon /var/swapfile
```

```
thenmozhi@thenmozhiVM:~/Desktop$ sudo fallocate -l 2G /var/swapfile
```

```
thenmozhi@thenmozhiVM:~/Desktop$ sudo chmod 600 /var/swapfile
```

```
thenmozhi@thenmozhiVM:~/Desktop$ sudo mkswap /var/swapfile
```

```
Setting up swapspace version 1, size = 2 GiB (2147479552 bytes)
```

```
no label, UUID=d5ce9c88-6d90-437f-8db9-0004e22843f7
```

```
thenmozhi@thenmozhiVM:~/Desktop$ sudo swapon /var/swapfile
```

```
thenmozhi@thenmozhiVM:~/Desktop$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	2.4Gi	782Mi	489Mi	9.0Mi	1.2Gi	1.5Gi
Swap:	3.1Gi	0B	3.1Gi			

Freeing Virtual Memory

1. Deactivate the swap file

```
sudo swapoff /var/swapfile
```

2. Remove the swapfile

```
sudo rm /var/swapfile
```

```
thenmozhi@thenmozhiVM:~/Desktop$ sudo swapoff /var/swapfile
thenmozhi@thenmozhiVM:~/Desktop$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	2.4Gi	805Mi	466Mi	9.0Mi	1.2Gi	1.5Gi
Swap:	1.1Gi	0B	1.1Gi			

```
thenmozhi@thenmozhiVM:~/Desktop$ sudo rm /var/swapfile
thenmozhi@thenmozhiVM:~/Desktop$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	2.4Gi	806Mi	466Mi	9.0Mi	1.2Gi	1.5Gi
Swap:	1.1Gi	0B	1.1Gi			



THANK YOU

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